a significant number of questions about small visual concepts, and textual answers are harder for MLLMs to guess from other side information in the image (compared to object types and attributes).

Sensitivity Study. If a model's perception is not sensitive to the size of visual concepts, we expect it to have similar accuracy in all three partitions. In Tab. 1, we observe that the accuracy of all MLLMs declines as the ground-truth bounding box becomes relatively smaller (right to left on the *no cropping* rows). BLIP-2 and InstructBLIP are not trained on TextVQA (*i.e.*, are zero-shot models), and their accuracy declines by 24 and 23 absolute percentage points between the `large` and `small` partitions, respectively. LLaVA-1.5 and Qwen-VL are trained on the training set of TextVQA, yet, their accuracy also declines by 11 and 12 between the `large` and `small` partitions, respectively. Lastly, even the most recent commercial GPT-4o, with an unknown training set that might include all of TextVQA, is suffering from a 7 percentage point decline in accuracy between small and medium partitions. These findings suggest that MLLMs have a bias against perceiving smaller visual concepts.

Intervention Study. The perceptual limitation we observed above might be merely correlated with size. To study whether this limitation is causally related to size, we conduct an intervention study where we provide the MLLMs with visually cropped images based on the ground-truth bounding boxes, denoted as `human-CROP`. More specifically, for each image-question pair and each MLLM, we crop the smallest square-shaped region containing the ground-truth bounding box from the image, and resize it to the input image resolution of the MLLM (the square-shaped cropping prevents extreme deformation of the cropped image when resizing). The cropped image is then provided to the MLLM in addition to the original image-question pair (see more details in Fig. 4). We observe in Tab. 1 that `human-CROP` significantly improves the accuracy of all MLLMs on the `small` and `medium` partitions, and to a lesser extent on the `large` partition. These findings show that the perception limitation is indeed caused by the size of the visual concepts, and that visual cropping can be a promising direction to mitigate this limitation.

4 Do MLLMs Know Where to Look?

The limitation in perceiving small visual concepts can have two primary reasons: 1) they are hard to locate in the larger image, and 2) their small details are hard to perceive correctly. In Fig. 1, we observed that the MLLM's incorrect answer may contain partially correct information, hinting that it might know where to look in the image. In this section, we quantitatively study that observation to answer whether MLLMs' sensitivity to size is rooted in a perception limitation or a localization limitation. To that end, we first utilize the attention maps computed inside the Transformer layers of an MLLM to quantify its spatial attention over the image and then compare the total amount of this attention inside the ground-truth bounding box to other bounding boxes of the same size.

MLLMs' Setup. The considered MLLMs process a given image-question pair (x, q) in four steps (depicted in Fig. 4): 1) the image is divided into $N \times N$ non-overlapping patches and processed by the ViT image encoder into $N \times N$ output tokens; 2) the ViT output tokens are transformed into the input space of the backbone LLM—by either an MLP (LLaVA-1.5) or a Transformer connector (BLIP-2, InstructBLIP and Qwen-VL)—which we refer to as image tokens; 3) the image tokens are then prepended to the question tokens and a predefined starting answer token, and fed to the LLM; 4) the LLM is sampled auto-regressively following the starting answer token (we use greedy sampling).

Quantifying MLLMs' Spatial Attention over the Image. We first measure how important each image token is to the MLLM's decision (*answer-to-token attention*) by extracting the softmax cross-attention of the starting answer token to all image tokens in all layers of the backbone LLM, resulting in $A_{st}(x, q) \in \mathbb{R}^{L \times H \times 1 \times T}$, where L, H are the number of layers and heads-per-layer in the LLM, and T is the number of image tokens provided to the LLM. We then take the average over attention heads to arrive at the answer-to-token attention $\hat{A}_{st}(x, q) = \frac{1}{H} \sum_{h=1}^{H} A_{st}(x, q)$. Next, we measure how important each image region is to each image token (*token-to-image attention*). For the MLLMs that use a Transformer connector to resample ViT output tokens into a fixed number of image tokens (BLIP-2, InstructBLIP and Qwen-VL), we extract the softmax cross-attention of each image token to all ViT output tokens in all layers of the connector, resulting in $A_{ti} \in \mathbb{R}^{L_c \times H_c \times T \times N^2}$, where L_c, H_c are the number of layers and heads-per-layer in the connector, T the number of learnable query tokens (that become input image tokens to the LLM), and N^2 the number of image patches of the ViT image encoder. We then take the average over attention heads to arrive at the token-to-image attention

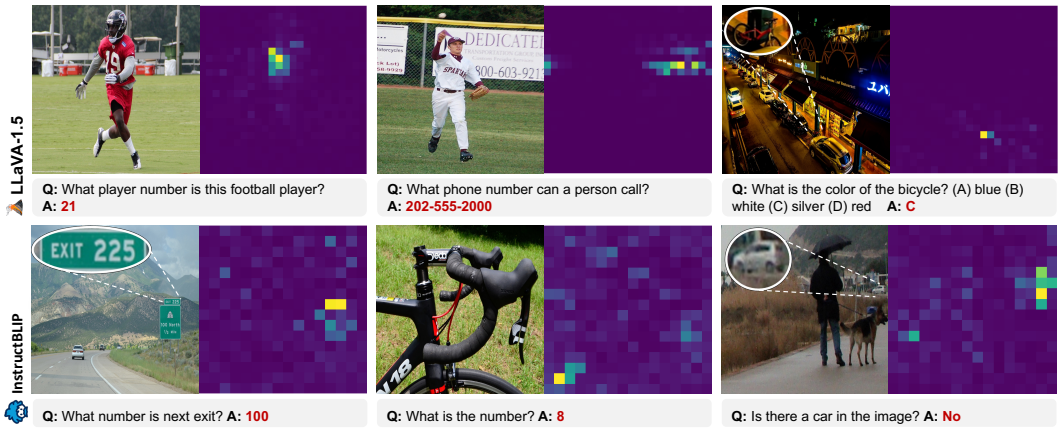

Figure 2: Examples of MLLMs knowing where to look despite answering incorrectly. The right panel in each example displays relative attention to the image (defined in Sec. 4) of one layer in the MLLM.

$\hat{A}_{ti}(x) = \frac{1}{H_c}\sum_{h=1}^{H_c} A_{ti}(x)$. For LLaVA-1.5 which uses an MLP to transform ViT output tokens to image tokens, we set $\hat{A}_{ti}(x)$ to the identity tensor. Finally, we compute the *answer-to-image attention* by computing the tensor product of the answer-to-token and token-to-image attention, resulting in $A_{si}(x,q) \in \mathbb{R}^{L\times L_c\times 1\times N^2}$ where $A_{si}^{mk}(x,q) = \hat{A}_{st}^{m}(x,q)\hat{A}_{ti}^{k}(x)$ (superscripts m and k denote layer indices on the LLM and the connector, respectively).

Relative Attention. One issue with using the softmax cross-attention is that not all highly attended tokens are semantically relevant to the input question. For example, recent work has observed that Transformers may use several tokens as registers to aggregate global information (Darcet et al., 2023). To emphasize semantically relevant attention, we propose to normalize the answer-to-image attention of an image-question pair (x, q) by its value on a generic instruction q'. Specifically, we consider a fixed instruction q' ="Write a general description of the image.", and compute **relative attention** as $A_{rel}(x,q) = \frac{A_{si}(x,q)}{A_{si}(x,q')}$ under element-wise division. Fig. 2 shows examples of relative attention for LLaVA-1.5 and InstructBLIP (A_{rel}^{mk} at layers $m = 14, k = 0$ and $m = 15, k = 2$, respectively).

Do MLLMs Know Where to Look? Equipped with relative attention, we now return to our question of whether MLLMs have a localization limitation or perception limitation. To that end, we consider the validation set of TextVQA again. For each image-question pair, we first compute the relative attention. We then define **attention ratio** as the ratio of the total (sum) relative attention inside the answer ground-truth bounding box to its average across all bounding boxes of the same size

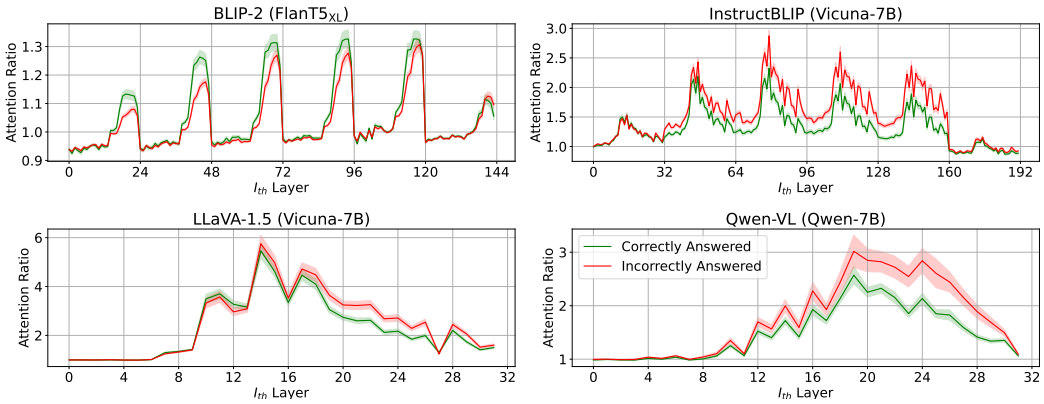

Figure 3: MLLMs' attention ratio across all layers (average with 95% CI over TextVQA). The attention ratio measures how significantly the MLLM is attending to the ground-truth bounding box (defined in Sec. 4). We observe that it is greater than 1 in most layers, showing that the MLLMs know where to look in the image even when they fail to answer correctly.

as the ground-truth on the image. This ratio provides a measure of how significantly the MLLM is attending to the ground-truth bounding box (in the sense of Markov's inequality). In Fig. 3, we plot the average (with 95% confidence interval) of the attention ratio, over the validation set of TextVQA for all layers in the considered MLLMs. The horizontal axis shows the combined layer index $l = m + k \times L$ for $m \in \{0 \dots L-1\}$ spanning the number of cross-attention layers in the backbone LLM, and $k \in \{0 \dots L_c - 1\}$ spanning the number of cross-attention layers in the connector (BLIP-2: $L = 24, L_c = 6$; InstructBLIP: $L = 32, L_c = 6$; Qwen-VL: $L = 32, L_c = 1$; LLaVA-1.5: $L = 32, L_c = 1$). In all MLLMs, we observe a significantly larger than 1 attention ratio in most layers, suggesting that the models are attending significantly to the ground-truth bounding box region on the image. Intriguingly, the models show similarly strong attention to the correct region regardless of whether they can answer the question correctly or incorrectly. These observations show that the MLLMs tend to know where to look, even when they answer incorrectly.

5 Automatic Visual Cropping (ViCrop)

We observed in Sec. 4 that the sensitivity of MLLMs to visual concept size is primarily a perception limitation (rather than a localization limitation). Therefore, one solution to mitigate this limitation is to simply train MLLMs with a larger number of image patches while maintaining per-patch resolution (hence increasing the image resolution of MLLMs). However, increasing the input image resolution by a factor of α, increases the number of ViT input patches (and output tokens) from N^2 to $\alpha^2 N^2$, which in turn increases the softmax attention computation complexity on the order of $\alpha^4 N^4$. Given that this solution is not scalable for current Transformer-based MLLMs, we choose to explore an alternative solution that **does not require any training and is scalable to any image resolution**. We note that several concurrent works have explored the first direction of *training* MLLMs with higher resolution image patches (Li et al., 2024c; Sun et al., 2024; Li et al., 2024d; McKinzie et al., 2024; Xu et al., 2024a; Luo et al., 2024), and notably LLaVA-Next (Liu et al., 2024a) has achieved the VQA state-of-the-art in several VQA benchmarks at the time of writing. We believe our work is parallel to these efforts in the following sense: rather than training higher and higher resolution MLLMs to enable them to see all resolutions (which is inevitably upper bounded), we explore how to smartly adjust the input image towards what an MLLM already can see without any additional training. We provide evidence showing that our training-free method can provide orthogonal benefits to the training-based methods in Appendices D and E.

Inspired by our findings that MLLMs tend to know where to look (Sec. 4) and that visual cropping can mitigate the perception limitation (Sec. 3), in this section we construct three automatic visual cropping methods in order to mitigate the perception limitation of MLLMs. These methods seek to use the internal information of an MLLM itself—in the form of attention maps and gradients—to find the approximate region of interest in images (*i.e.*, the region containing the subject of a question), and then to zoom into that region via visual cropping. One potential drawback of visual cropping is that some questions might need to have a global view of the image. To address this issue, we utilize the fact that MLLMs typically convert the image into a series of tokens. This allows us to directly

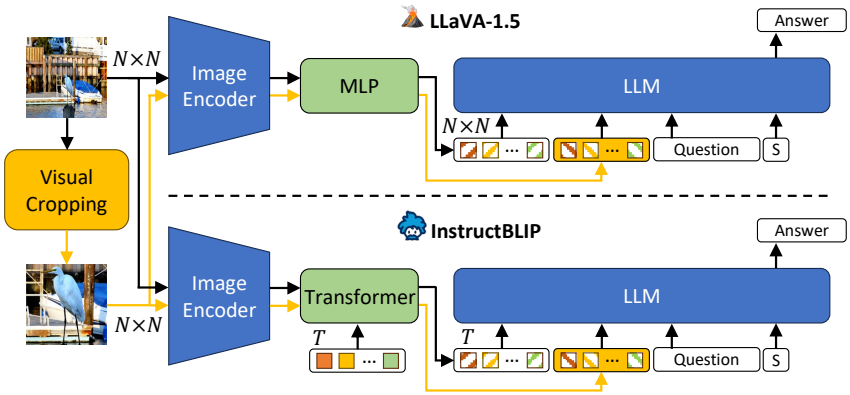

Figure 4: Illustration of the proposed visual cropping approach applied to two MLLMs.